西北农林科技大学
NORTHWEST A&F UNIVERSITY

图 像 处 理 与 机 器 视 觉

课后作业

作业题目：第五次作业

学　　院：机械与电子工程学院

专业班级：机化 2201

姓　　名：黄钊伟

学　　号：2022013106

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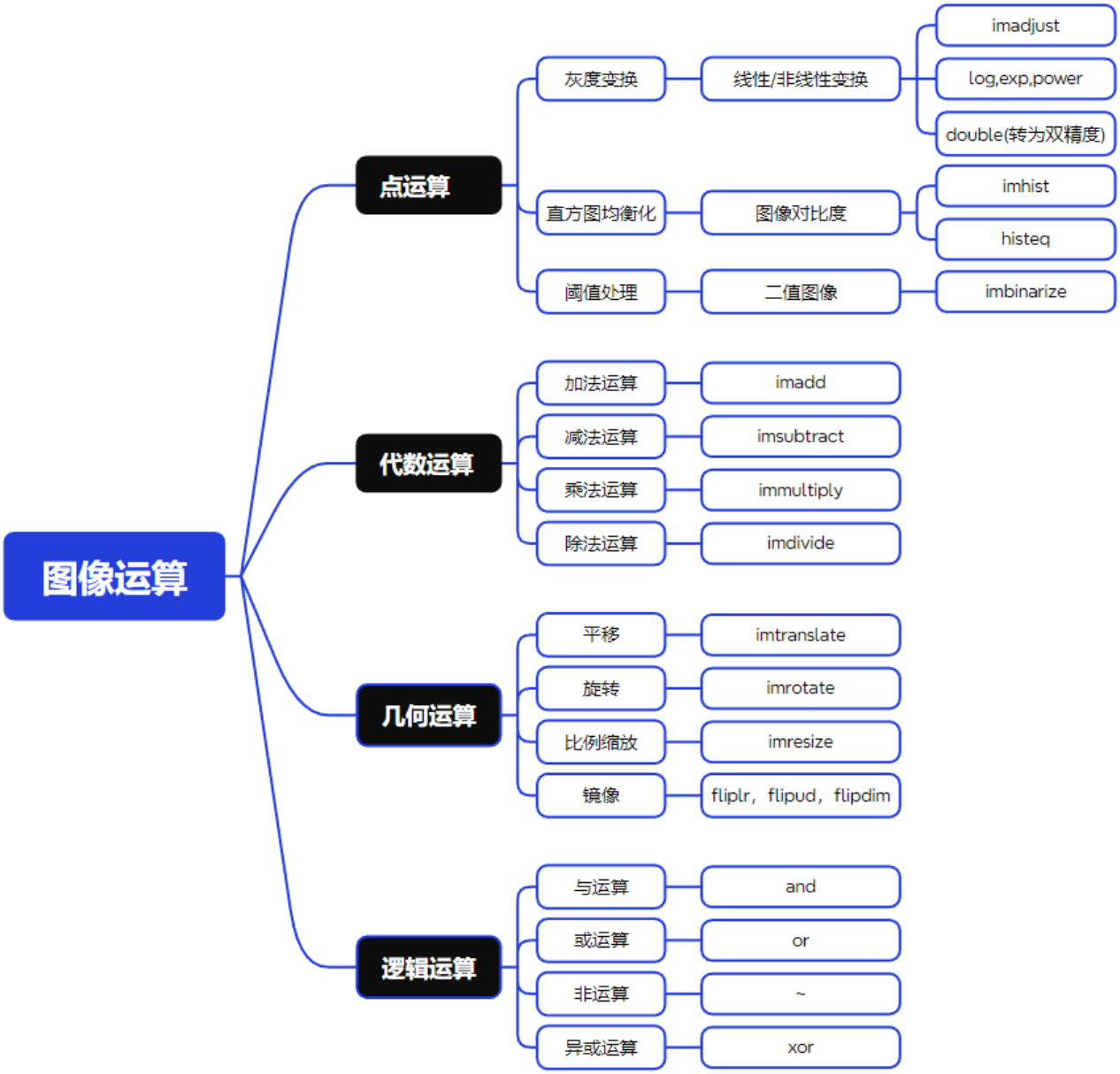
西北农林科技大学

机械与电子工程学院

1、整理图像变换的四种运算的具体分类及常用代码。



2、在人像周围插入几只小蝴蝶，插入时旋转不同角度。

运行代码：

（有点难度，向 AI 请教了一下，醍醐灌顶）

youle_20240923.m　youle_20240926.m　youle_20240930.m

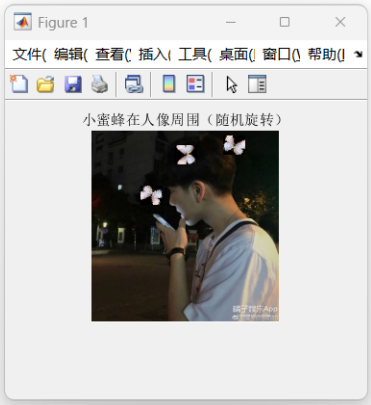
```
fly = imread('fly.png');
head = imread('Head.png');
suoxiao = 0.15;
Image1 = imresize(fly, suoxiao);
background = (Image1(:,:,1) == 0) & (Image1(:,:,2) == 0) & (Image1(:,:,3) == 0);
Image1(repmat(background, [1, 1, 3])) = NaN;
number = 3;
Image2 = head;
[fly1, fly2, ~] = size(Image1);

for i = 1:number
    x = randi([1, size(head, 2) - fly2]);
    y = randi([1, size(head, 1) - fly1]);
    angle = randi([0, 360]);
    rotatedImage = imrotate(Image1, angle, 'crop');
    [rotatedImage1, rotatedImage2, ~] = size(rotatedImage);

    if x + rotatedImage2 > size(head, 2)
        x = size(head, 2) - rotatedImage2;
    end
    if y + rotatedImage1 > size(head, 1)
        y = size(head, 1) - rotatedImage1;
    end

    for c = 1:3
        region1 = rotatedImage(:,:,c);
```

youle_20240923.m　youle_20240926.m　youle_20240930.m

```
Image2 = head;
[fly1, fly2, ~] = size(Image1);

for i = 1:number
    x = randi([1, size(head, 2) - fly2]);
    y = randi([1, size(head, 1) - fly1]);
    angle = randi([0, 360]);
    rotatedImage = imrotate(Image1, angle, 'crop');
    [rotatedImage1, rotatedImage2, ~] = size(rotatedImage);

    if x + rotatedImage2 > size(head, 2)
        x = size(head, 2) - rotatedImage2;
    end
    if y + rotatedImage1 > size(head, 1)
        y = size(head, 1) - rotatedImage1;
    end

    for c = 1:3
        region1 = rotatedImage(:,:,c);
        region2 = Image2(y:y+rotatedImage1-1, x:x+rotatedImage2-1, c);
        region2(~isnan(region1)) = region1(~isnan(region1));
        Image2(y:y+rotatedImage1-1, x:x+rotatedImage2-1, c) = region2;
    end
end

imshow(Image2);
title('Handsome');
```

运行结果：

```
Image2 = head;
[fly1, fly2, ~] = size(Image1);

for i = 1:number
    x = randi([1, size(head, 2) - fly2]);
    y = randi([1, size(head, 1) - fly1]);
    angle = randi([0, 360]);
    rotatedImage = imrotate(Image1, angle, 'crop');
    [rotatedImage1, rotatedImage2, ~] = size(rotatedImage);

    if x + rotatedImage2 > size(head, 2)
        x = size(head, 2) - rotatedImage2;
    end
    if y + rotatedImage1 > size(head, 1)
        y = size(head, 1) - rotatedImage1;
    end

    for c = 1:3
        region1 = rotatedImage(:,:,c);
        region2 = Image2(y:y+rotatedImage1-1, x:x+rotatedImage2-1, c);
        region2(~isnan(region1)) = region1(~isnan(region1));
        Image2(y:y+rotatedImage1-1, x:x+rotatedImage2-1, c) = region2;
    end
end

imshow(Image2);
title('Handsome');
```



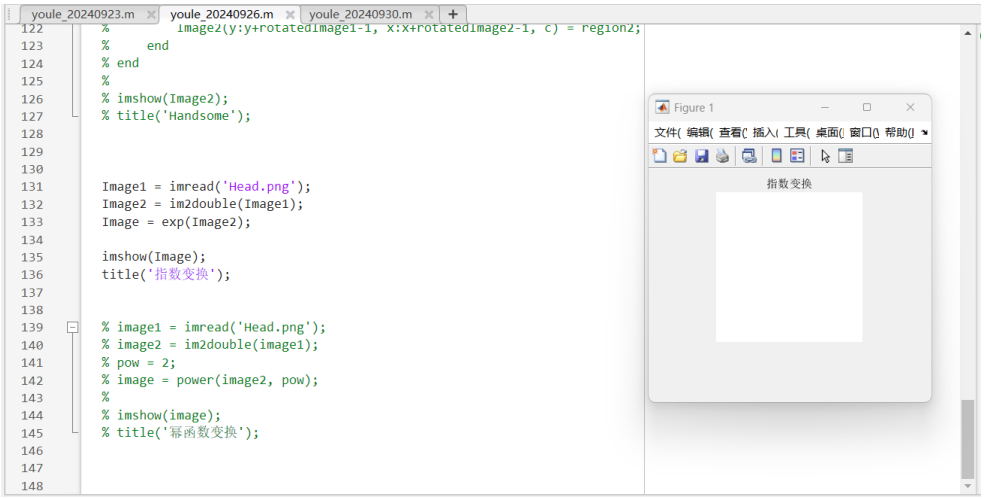
3、分别对图像进行指数函数变换和幂函数变换。

①指数函数变换：

运行代码：

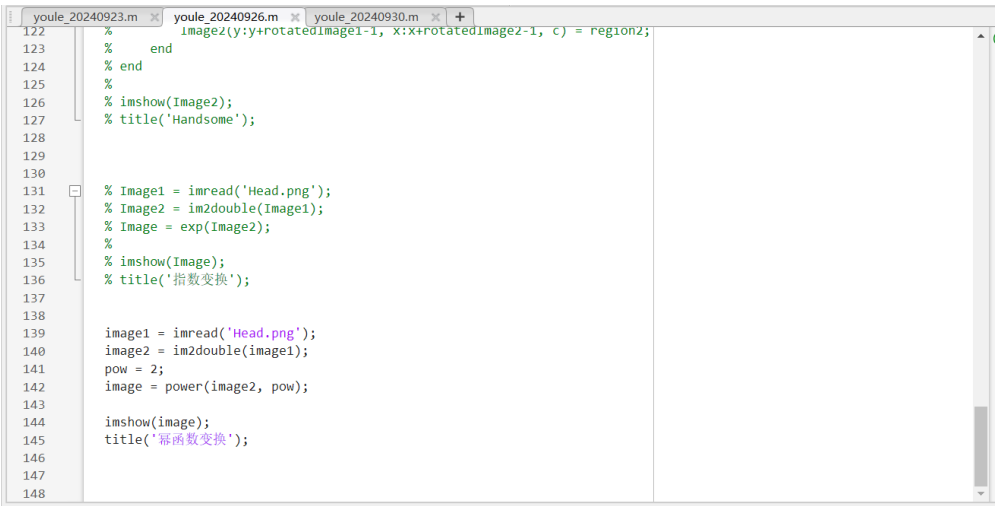
```
%         image2(y:y+rotatedImage1-1, x:x+rotatedImage2-1, c) = region2;
%     end
% end
%
% imshow(Image2);
% title('Handsome');



Image1 = imread('Head.png');
Image2 = im2double(Image1);
Image = exp(Image2);

imshow(Image);
title('指数变换');


% image1 = imread('Head.png');
% image2 = im2double(image1);
% pow = 2;
% image = power(image2, pow);
%
% imshow(image);
% title('幂函数变换');
```

运行结果：（变换完看不见了）

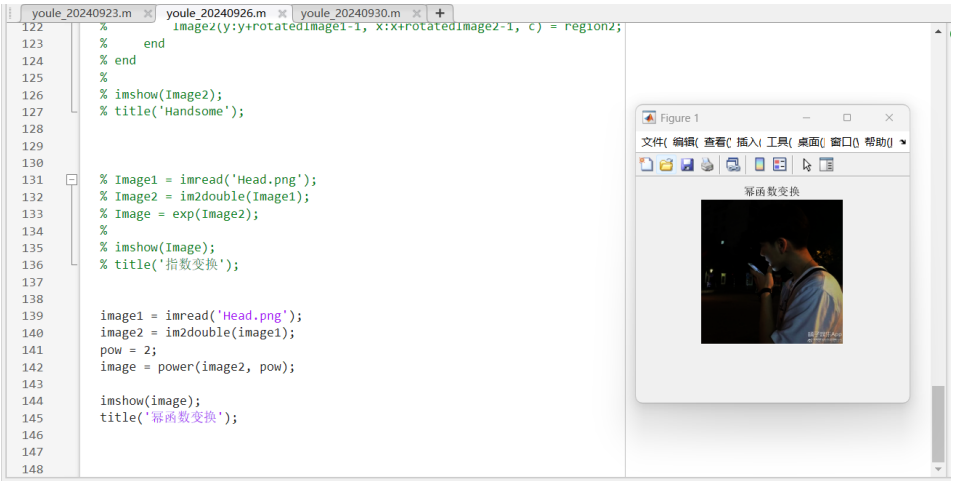


②幂函数变换：

运行代码：



运行结果：



4、简述直方图均衡化的基本原理。

直方图均衡化是一种图像处理技术，用于增强图像的对比度，它通过调整图像的灰度直方图，使得输出图像的直方图分布更加均匀，从而在不同的灰度级别上均匀分布像素。该方法首先计算图像的灰度直方图，然后利用累积分布函数（CDF）来设计一个新的灰度级映射，使得每个灰度级的像素数量在均衡化后的图像中均匀分布。具体来说，它将 CDF 值线性映射到新的灰度级，然后使用这个映射替换原始图像中的每个像素值，以生成新的均衡化图像。